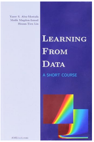
MSML610: Advanced Machine Learning

# Lesson 05.1: Machine Learning Theories

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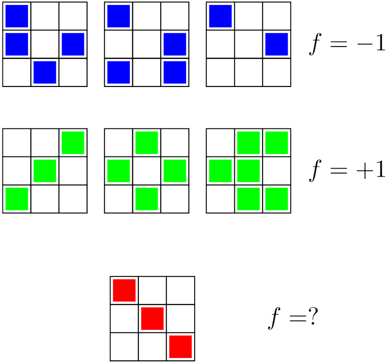
**References**:

- Abu-Mostafa et al.: *"Learning From Data"* (2012)

- ***Is Machine Learning Even Possible?***
- Growth Function
- The VC Dimension

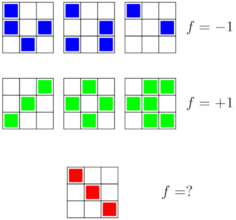
# A Simple Visual ML Experiment (1/2)

- Consider the supervised classification problem
- **Input**
  - A 9 bit vector represented as a 3x3 array
- **Training set**
  - The blue row $\underline{\boldsymbol{x}}_1, \underline{\boldsymbol{x}}_2, \underline{\boldsymbol{x}}_3$ for $f(\underline{\boldsymbol{x}}) = -1$
  - The green row $\underline{\boldsymbol{x}}_4, \underline{\boldsymbol{x}}_5, \underline{\boldsymbol{x}}_6$ for $f(\underline{\boldsymbol{x}}) = +1$
- **Test set**
  - For the red pattern $\underline{\boldsymbol{x}}_0$, is $f(\underline{\boldsymbol{x}}_0) = -1$ or $+1$?



# A Simple Visual ML Experiment (2/2)

- **Model 1**
  - $f(\underline{\boldsymbol{x}}) = +1$ when $\underline{\boldsymbol{x}}$ has an axis of symmetry
  - $f(\underline{\boldsymbol{x}}) = -1$ when $\underline{\boldsymbol{x}}$ is not symmetric
  - The test set is symmetrical $\implies f(\underline{\boldsymbol{x}}_0) = +1$
- **Model 2**
  - $f(\underline{\boldsymbol{x}}) = +1$ when the top left square $\underline{\boldsymbol{x}}$ is empty
  - $f(\underline{\boldsymbol{x}}) = -1$ when the top left square $\underline{\boldsymbol{x}}$ is full
  - The test set has top left square full $\implies f(\underline{\boldsymbol{x}}_0) = -1$
- Many functions fit the 6 training examples
  - Some have a value of -1 on the test point, others $+1$
  - Which one is it?
- How can a limited data set reveal enough information to define the entire target function?
  - **Is machine learning possible?**

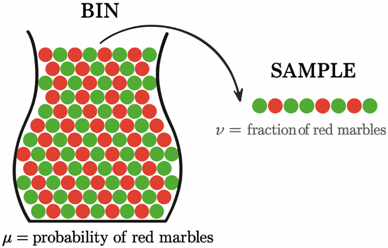


# Is Machine Learning Possible?

- The function can assume **any value outside data**
  - E.g., with summer temperature data, the function could assume a different value for winter
- **How to learn an unknown function?**
  - Estimating at unseen points seems impossible in general
  - Requires assumptions or models about behavior
- Difference between:
  - **Possible**
    - No knowledge of the unknown function
    - E.g., could be linear, quadratic, or sine wave outside known data
  - **Probable**
    - Some knowledge of the unknown function from domain knowledge or historical data patterns
    - E.g., if historical weather data forms a sinusoidal pattern, unknown points likely follow that pattern

# Supervised Learning: Bin Analogy (1/2)

- Consider a bin with red and green marbles
  - We want to estimate $\Pr(\text{pick a red marble}) = \mu$ where the value of $\mu$ is unknown
  - We pick $N$ marbles independently with replacement
  - The fraction of red marbles is $\nu$




- Does $\nu$ say anything about $\mu$?
  - **"No"**
    - In strict terms, we don't know anything about the marbles we didn't pick
    - The sample can be mostly green, while the bin is mostly red
    - This is *possible*, but *not probable*
  - **"Yes"**
    - Under certain conditions, the sample frequency is close to the real frequency

# Hoeffding Inequality

- Consider a Bernoulli random variable $X$ with probability of success $\mu$
- Estimate the mean $\mu$ using $N$ samples with $\nu = \frac{1}{N} \sum_i X_i$
- The **probably approximately correct** (PAC) statement holds:

$$\Pr(|\nu - \mu| > \varepsilon) \leq \frac{2}{e^{2\varepsilon^2 N}}$$

- **Remarks:**
  - Valid for all $N$ and $\varepsilon$, not an asymptotic result
  - Holds only if you sample $\nu$ and $\mu$ at random and in the same way
  - If $N$ increases, it is exponentially small that $\nu$ will deviate from $\mu$ by more than $\varepsilon$
  - The bound does not depend on $\mu$
  - Trade-off between $N$, $\varepsilon$, and the bound:
    * Smaller $\varepsilon$ requires larger $N$ for the same probability bound
    * Since $\nu \in [\mu - \varepsilon, \mu + \varepsilon]$, you want small $\varepsilon$ with a large probability
  - It is a statement about $\nu$ and not $\mu$ although you use it to state something about $\nu$ (like for a confidence interval)

# Supervised Learning: Bin Analogy (2/2)

- Let's connect the bin analogy, Hoeffding inequality, and feasibility of machine learning
  - You know $f(\underline{\boldsymbol{x}})$ at points $\underline{\boldsymbol{x}} \in \mathcal{X}$
  - You choose an hypothesis $h : \mathcal{X} \rightarrow \mathcal{Y} = \{0, 1\}$
  - Each point $\underline{\boldsymbol{x}} \in \mathcal{X}$ is a marble
  - You color red if the hypothesis is correct $h(\underline{\boldsymbol{x}}) = f(\underline{\boldsymbol{x}})$, green otherwise
  - The in-sample error $E_{in}(h)$ corresponds to $\nu$
  - The marbles of unknown color corresponds to $E_{out}(h) = \mu$
  - $\underline{\boldsymbol{x}}_1, ..., \underline{\boldsymbol{x}}_n$ are picked randomly and independently from a distribution over $\mathcal{X}$ which is the same as for $E_{out}$
- Hoeffding inequality holds and bounds the error going from in-sample to out-of-sample

$$\Pr(|E_{in} - E_{out}| > \varepsilon) \leq c$$

  - Generalization over unknown points (i.e., marbles) is possible
  - **Machine learning is possible!**

# Validation vs Learning: Bin Analogy

- You have learned that for a given $h$, in-sample performance $E_{in}(h) = \nu$ needs to be close to out-of-sample performance $E_{out}(h) = \mu$
  - This is the **validation setup**, after you have already learned a model
- In a **learning setup** you have $h$ to choose from $M$ hypotheses
  - You need a bound on the out-of-sample performance of the chosen hypothesis $h \in \mathcal{H}$, regardless of which hypothesis you choose
  - You need a Hoeffding counterpart for the case of choosing from multiple hypotheses

$$
\begin{aligned}
\forall g \in \mathcal{H} = \{h_1, \dots, h_M\} \ & \Pr(|E_{in}(g) - E_{out}(g)| > \varepsilon) \\
&\leq \Pr\Big(\bigcup_{i=1}^{M} (|E_{in}(h_i) - E_{out}(h_i)| > \varepsilon)\Big) \\
&\leq \sum_{i=1}^{M} \Pr(|E_{in}(h_i) - E_{out}(h_i)| > \varepsilon) && \text{(by the union bound)} \\
&\leq 2M \exp(-2\varepsilon^2 N) && \text{(by Hoeffding)}
\end{aligned}
$$

- **Problem**: the bound is weak

# Validation vs Learning: Coin Analogy

- In a **validation set-up**, you have a coin and want to determine if it is fair
- Assume the coin is unbiased: $\mu = 0.5$
- Toss the coin 10 times
- How likely is that you get 10 heads (i.e., the coin looks biased $\nu = 0$)?

$$\Pr(\text{coin shows } \nu = 0) = 1/2^{10} = 1/1024 \approx 0.1\%$$

- **Conclusion**: the probability that the out-of-sample performance ($\nu = 0.0$) is completely different from the in-sample perf ($\mu = 0.5$) is very low

# Validation vs Learning: Coin Analogy

- In a **learning set-up**, you have many coins and you need to choose one and determine if it's fair
- If you have 1000 fair coins, how likely is it that at least one appears totally biased using 10 experiments?
  - I.e., out-of-sample performance is completely different from in-sample performance

$$\begin{aligned}
\text{Pr}(\text{at least one coin has } \nu = 0) &= 1 - \text{Pr}(\text{all coins have } \nu \neq 0) \\
&= 1 - (\text{Pr}(\text{a coin has } \nu \neq 0))^{10} \\
&= 1 - (1 - \text{Pr}(\text{a coin has } \nu = 0))^{10} \\
&= 1 - (1 - 1/2^{10})^{1000} \\
&\approx 0.63\%
\end{aligned}$$

- **Conclusion**: It is probable, more than 50%

# Validation vs Learning: Hoeffding Inequality

- In **validation / testing**
  - Use Hoeffding to assess how well our $g$ (the *chosen hypothesis*) approximates $f$ (the *unknown hypothesis*):

    $$\Pr(|E_{in} - E_{out}| > \varepsilon) \leq 2\exp(-2\varepsilon^2 N)$$

    where:

    $$E_{in}(g) = \frac{1}{N}\sum_i e(g(\underline{\boldsymbol{x}}_i), f(\underline{\boldsymbol{x}}_i))$$

    $$E_{out}(g) = \mathbb{E}_{\underline{\boldsymbol{x}}}[e(g(\underline{\boldsymbol{x}}), f(\underline{\boldsymbol{x}}))]$$

  - Since the hypothesis $g$ is final and fixed, Hoeffding inequality guarantees that you can learn since it gives a bound for $E_{out}$ to track $E_{in}$
- In **learning**
  - Need to account that our hypothesis is the best of $M$ hypotheses, so:

    $$\Pr(|E_{in} - E_{out}| > \varepsilon) \leq 2M\exp(-2\varepsilon^2 N)$$

  - The bound for $E_{out}$ from Hoeffding is weak
- **Questions**:

# Intuition Why Bound for Hoeffding Is Weak

- The Hoeffding inequality and the union bound applied to training set

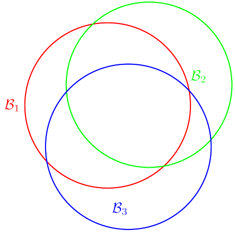
$$\Pr(|E_{in} - E_{out}| > \varepsilon) \leq 2M \exp(-2\varepsilon^2 N)$$

  is **artificially** too loose

- $M$ was coming from the bad event:

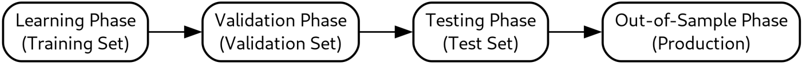
$$\begin{aligned}\mathcal{B}_i &= \textit{"hypothesis } h_i \textit{ does not generalize out-of-sample"} \\ &= \text{"}|E_{in}(h_i) - E_{out}(h_i)| > \varepsilon\text{"}\end{aligned}$$

- Since $g \in \{h_1, h_2, \cdots, h_M\}$ then $\Pr(\mathcal{B}) \leq \Pr(\bigcup_i \mathcal{B}_i) \leq \sum_i \Pr(\mathcal{B}_i)$
- The union bound assumes the events are disjoint, leading to a conservative estimate if events overlap
- **In reality**, bad events are extremely overlapping because bad hypotheses are extremely similar



# Training vs Testing: College Course Analogy (1/2)

- In machine learning there are several phases




- This set-up is very similar to studying and exams in a college course
- Students study the material
  - This is the **learning phase**
- Before the final exam, students receive practice problems and solutions
  - Studying the problems improves performance by understating what they need to improve
  - This corresponds to the **validation set**

# Training vs Testing: College Course Analogy (2/2)

- The final exam corresponds to the **testing phase**
  - These problems are different than the problems in the validation set
  - Why not give out exam problems to improve performance?
    - Doing well in the exam isn't the goal
    - The goal is to learn the course material
  - The final exam isn't strictly necessary
    - Gauges how well you've learned
    - Motivates you to study
    - Knowing exam problems in advance wouldn't gauge learning effectively
- What matters is how students do once they graduate and find a job
  - This is the **out-of-sample phase**

- Is Machine Learning Even Possible?
- ***Growth Function***
- The VC Dimension

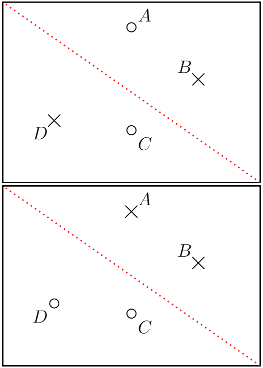
# Dichotomy: Definition

- **Problem**: classify $N$ (fixed) points $\underline{\boldsymbol{x}}_1, \ldots, \underline{\boldsymbol{x}}_N$ with an hypothesis set $\mathcal{H}$ of multi-class classifiers
- Consider an assignment $D$ of the points to certain class $\underline{\boldsymbol{d}}_1, \ldots, \underline{\boldsymbol{d}}_N$
- $D$ is a **dichotomy** for hypothesis set $\mathcal{H}$ $\iff$ there exists $h \in \mathcal{H}$ that gets the desired classification $D$

- **Example**
  - 4 points in a plane $A, B, C, D$
  - Binary classification
  - $\mathcal{H} = \{$ bidimensional perceptrons $\}$
  - Moving the separating hyperplane, you get different classifications for the points (i.e., dichotomies)

| | D1 | D2 | D3 | D4 | D... |
|---|---|---|---|---|---|
| A | o | x | ... | | |
| B | x | x | | | |
| C | o | o | | | |
| D | x | o | ... | | |

  - There are at most $2^N$ dichotomies
  - Certain classifications are not possible



# Dichotomies vs Hypotheses

- An **hypothesis** classifies each point of $\mathcal{X}$: $\mathcal{X} \to \{-1, +1\}$
- A **dichotomy** classifies each point of a fixed set: $\{\underline{\boldsymbol{x}}_1, ..., \underline{\boldsymbol{x}}_N\} \to \{-1, +1\}$
  - Dichotomies are "mini-hypotheses", i.e., hypotheses restricted to given points
  - A dichotomy depends on:
    * The number of points $N$
    * Hypothesis set $\mathcal{H}$ (i.e., the possible models)
    * Where the points are placed
    * How the points are assigned
- The **number of different dichotomies** is indicated by $|\mathcal{H}(\underline{\boldsymbol{x}}_1, ..., \underline{\boldsymbol{x}}_N)|$
  - The number of dichotomies is always finite, since $|\mathcal{H}(\underline{\boldsymbol{x}}_1, ..., \underline{\boldsymbol{x}}_N)| \leq N^K$
  - The number of hypotheses is usually infinite, i.e., $|\mathcal{H}| = \infty$
- The "complexity" of $\mathcal{H}$ is related to the number of hypothesis
- From the training set point of view what matters are dichotomies and not hypotheses

# Growth Function

- The **growth function** counts the maximum number of possible dichotomies on $N$ points for a hypothesis set $\mathcal{H}$:

$$m_{\mathcal{H}}(N) = \max_{\underline{\boldsymbol{x}}_1, \cdots, \underline{\boldsymbol{x}}_N \in \mathcal{X}} |\mathcal{H}(\underline{\boldsymbol{x}}_1, \cdots, \underline{\boldsymbol{x}}_N)|$$

- **Why growth function?**
  - The dichotomies depend on point distribution and assignment
  - The growth function considers the maximum by placing points in the most “favorable way” for the hypothesis set
- To compute $m_{\mathcal{H}}(N)$ by **brute force**:
  - Consider all possible placements of $N$ points $\underline{\boldsymbol{x}}_1, ..., \underline{\boldsymbol{x}}_N$
  - Consider all possible assignments of the points to the classes
  - Consider all possible hypotheses $h \in \mathcal{H}$
  - Compute the corresponding dichotomy for $h$ on $\underline{\boldsymbol{x}}_1, ..., \underline{\boldsymbol{x}}_N$
  - Count the number of different dichotomies

# What Can Vary in a Dichotomy

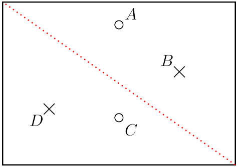
- Given:
  - An hypothesis set $\mathcal{H}$ (e.g., bidimensional perceptrons)
  - $N$ (fixed) points $\underline{\boldsymbol{x}}_1, \ldots, \underline{\boldsymbol{x}}_N$
  - An assignment $D$ of the points to certain class $\underline{\boldsymbol{d}}_1, \ldots, \underline{\boldsymbol{d}}_N$
- $D$ is a **dichotomy** for hypothesis set $\mathcal{H} \iff$ there exists $h \in \mathcal{H}$ that gets the desired classification $D$
- There are various quantities in the definition of dichotomy
  - The hypothesis set $\mathcal{H}$
    * It is fixed
  - The number of dimensions of the input space
    * It is fixed through the hypothesis set $\mathcal{H}$
  - The number of points $N$
    * Input to the growth function $m_{\mathcal{H}}(N)$
  - How the points are assigned to the classes $\underline{\boldsymbol{d}}_1, \ldots, \underline{\boldsymbol{d}}_N$
    * It is a free parameter, removed by how each hypothesis in $\mathcal{H}$ "splits" the space
  - Where the points are positioned $\underline{\boldsymbol{x}}_1, \ldots, \underline{\boldsymbol{x}}_N$
    * It is a free parameter, removed by the growth function through

# Growth Function Is Increasing

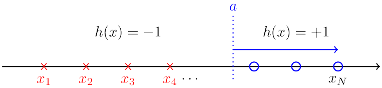
- $m_{\mathcal{H}}(N)$ increases (although not monotonically) with $N$
- E.g.,
  - The number of dichotomies on $N = 3$ points $m_{\mathcal{H}}(3)$ is smaller or equal than the number of dichotomies on $N = 4$ points
  - In fact we can ignore a new point and get the same classification
- $m_{\mathcal{H}}(N)$ increases with the complexity of $\mathcal{H}$
- $m_{\mathcal{H}}(N)$ increases with the number of dimensions in the input space (i.e., feature space)

## Growth Function: Examples

- Consider the growth function $m_{\mathcal{H}}$ for different hypothesis sets $\mathcal{H}$
- **Perceptron on a plane**
  - $m_{\mathcal{H}}(3) = 8$
  - $m_{\mathcal{H}}(4) = 14$ (2 XOR classifications not possible)

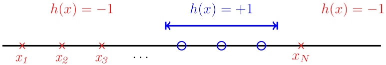



- **Positive rays** $\text{sign}(x - a)$ on $\mathbb{R}$
  - $m_{\mathcal{H}}(N) = N + 1$
  - Origin of rays $a$ can be placed in $N + 1$ intervals

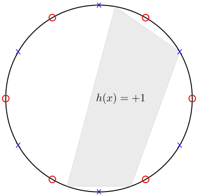


# Growth Function: Examples

- **Positive intervals** on $\mathbb{R}$ $x \in [a, b]$
  - $m_{\mathcal{H}}(N) = \binom{N+1}{2} + 1 \sim \mathrm{N}^2$
  - Pick 2 distinct intervals out of $N+1$, and there is a dichotomy with 2 points in the same interval




- **Convex sets on a plane**
  - $m_{\mathcal{H}}(N) = 2^N$
  - Place points in a circle and can classify $N$ points in any way



# Break Point of an Hypothesis Set

- Given an hypothesis set $\mathcal{H}$
- A hypothesis set $\mathcal{H}$ **shatters $N$ points** $\iff m_{\mathcal{H}}(N) = 2^N$
  - There is a position of $N$ points and a class assignment that you can classify using $h \in \mathcal{H}$
  - It does not mean all sets of $N$ points can be classified in any way
- $k$ is a **break point** for $\mathcal{H} \iff m_{\mathcal{H}}(k) < 2^k$
  - I.e., no data set of size $k$ can be shattered by $\mathcal{H}$
  - E.g.,
    - For 2D perceptron: a break point is 4
    - For positive rays: a break point is 2
    - For positive intervals: a break point is 3
    - For convex set on a plane: there is no break point

# Break Point for an Hypothesis Set and Learning

- If there is a break point for a hypothesis set $\mathcal{H}$, it can be shown that:
  - $m_{\mathcal{H}}(N)$ is polynomial in $N$
  - Instead of Hoeffding's inequality for learning

$$\Pr(|E_{in}(g) - E_{out}(g)| > \varepsilon) \leq 2Me^{-2\varepsilon^2 N}$$

    you can use the Vapnik-Chervonenkis inequality:

$$\Pr(\text{bad generalization}) \leq 4m_{\mathcal{H}}(2N)e^{-\frac{1}{8}\varepsilon^2 N}$$

  - Since $m_{\mathcal{H}}(N)$ is polynomial in $N$, it will be dominated by the negative exponential, given enough examples
  - You can have a generalization bound: machine learning works!
- A hypothesis set can be characterized from the learning point of view by the **existence and value of a break point**

- Is Machine Learning Even Possible?
- Growth Function
- ***The VC Dimension***

# VC Dimension of an Hypothesis Set

- The **VC dimension of a hypothesis set** $\mathcal{H}$, denoted as $d_{VC}(\mathcal{H})$, is defined as the largest value of $N$ for which $m_{\mathcal{H}}(N) = 2^N$
  - I.e., the VC dimension is the most points $\mathcal{H}$ can shatter
- **Properties** of the VC dimension: if $d_{VC}(\mathcal{H}) = N$ then
  - Exists a constellation of $N$ points that can be shattered by $\mathcal{H}$
    - Not all sets of $N$ points can be shattered
    - If $N$ points were placed randomly, they could not be necessarily shattered
  - $\mathcal{H}$ can *shatter* $N$ points for any $N \leq d_{VC}(\mathcal{H})$
  - The *smallest break point* is $d_{VC} - 1$
  - The *growth function* in terms of the VC dimension is $m_{\mathcal{H}} \leq \sum_{i=0}^{d_{VC}} \binom{N}{i}$
  - The VC dimension is the *order of the polynomial bounding $m_{\mathcal{H}}$*

# VC Dimension: Interpretation

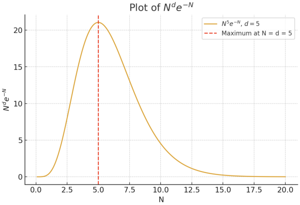
- The VC dimension **measures the complexity** of a hypothesis set in terms of **effective parameters**
- E.g.,
  - A perceptron in a $d$-dimensional space has $d_{VC} = d + 1$
  - In fact $d_{VC}$ is the number of perceptron parameters!
  - E.g., for a 2D perceptron ($d = 2$), the break point is 2, so $d_{VC} = 3$
- The VC dimension considers the model as a black box in order to estimate effective parameters
  - How many points $N$ a model can shatter, not the number of parameters
- Not all parameters contribute to degrees of freedom
  - E.g., combining $N$ 1D perceptrons gives $2N$ parameters, but the effective degrees of freedom remain 2
- A complex hypothesis $\mathcal{H}$:
  - Has more parameters (higher VC dimension $d_{VC}$)
  - Requires more examples for training

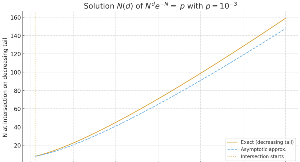
# VC Generalization Bounds

- How many data points are needed to obtain $\Pr(|E_{in} - E_{out}| > \varepsilon) \leq \delta$?
- The VC inequality states

$$\Pr(\text{bad generalization}) \leq 4 m_{\mathcal{H}}(2N) e^{-\frac{1}{8}\varepsilon^2 N}$$

- $N^d e^{-N}$ abstracts the upper bound term
  - Plot $N^d e^{-N}$ vs. $N$: Power dominates for small $N$, exponential for large $N$ and brings it to 0
  - Vary $d$ (VC dimension) function peaks for larger $N$, then approaches the region of interest $< 1$
- Plot intersection of $N^d e^{-N}$ with a probability as a function of $d$
  - Examples $N$ needed are proportional to $d$
  - Rule of thumb: $N > 10 d_{VC}$ for



# VC Generalization Bounds

- The VC inequality

$$\Pr(|E_{in} - E_{out}| > \varepsilon) \leq 4m_{\mathcal{H}}(2N)e^{-\frac{1}{8}\varepsilon^2 N}$$

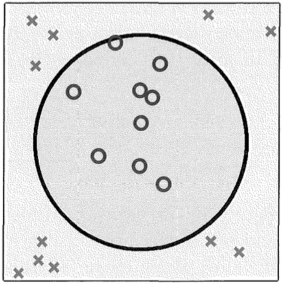
can be used in several ways to relate $\varepsilon$, $\delta$, and $N$, e.g.,

- Examples
  - "Given $\varepsilon = 1\%$ error, how many examples $N$ are needed to get $\delta = 0.05$?"
  - "Given $N$ examples, what's the probability of an error larger than $\varepsilon$?"
- You can equate $\delta$ to $4m_{\mathcal{H}}(2N)e^{\frac{1}{8}\varepsilon^2 N}$ and solve for $\varepsilon$, getting

$$\Omega(N, \mathcal{H}, \delta) = \sqrt{\frac{8}{N} \ln \frac{4m_{\mathcal{H}}(2N)}{\delta}}$$

- Then you can say $|E_{out} - E_{in}| \leq \Omega(N, \mathcal{H}, \delta)$ with probability $\geq 1 - \delta$
  - The generalization bounds are then: $\Pr(E_{out} \leq E_{in} + \Omega) \geq 1 - \delta$

# How to Void the VC Analysis Guarantee



- Consider the case where data is genuinely non-linear
  - E.g., “o” points in the center and “x” in the corners
- Transform to high-dimensional $\mathcal{Z}$ with:

$$\Phi : \underline{\boldsymbol{x}} = (x_0, ..., x_d) \to \underline{\boldsymbol{z}} = (z_0, ..., z_{\tilde{d}})$$

- $d_{VC} \leq \tilde{d} + 1$; smaller $\tilde{d}$ improves generalization
  - Use $\underline{\boldsymbol{z}} = (1, x_1, x_2, x_1 x_2, x_1^2, x_2^2)$
  - Why not $\underline{\boldsymbol{z}} = (1, x_1^2, x_2^2)$?
  - Why not $\underline{\boldsymbol{z}} = (1, x_1^2 + x_2^2)$?
  - Why not $\underline{\boldsymbol{z}} = (x_1^2 + x_2^2 - 0.6)$?
- Some model coefficients were zero and discarded, leaving machine learning the rest
  - VC analysis is a warranty, forfeited if data is examined before model selection (data snooping)
  - From VC analysis, complexity is that of the initial hypothesis set